Resolving the Hubble Tension Through Tetrad Topology

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Abstract

The Hubble tension describes the persistent discrepancy between early-universe measurements ($H \approx 67$ km/s/Mpc) and late-universe measurements ($H \approx 73$ km/s/Mpc). This paper applies the tetrad formalism of General Relativity to this problem. The tetrad requires $k = 4$ orthonormal basis vectors (one timelike, three spacelike) to close the local frame. This same $k = 4$ appears in the ADM constraint structure (1 Hamiltonian + 3 momentum), LIGO's polarization count ($10 - 2k = 2$), and Regge calculus. Field closure with $k = 4$ topology is the dimensional requirement of 3+1 spacetime.

Two independent routes identify the minimal closure size of the gravitational field: 2.82 fm. Electromagnetic stability gives $r_e = e^2/4\pi\varepsilon_0 m_e c^2 = 2.818$ fm; gravitational topology gives $r_e = k\sqrt{Gm_e/a_0} = 2.82$ fm. These routes use different physics yet converge to 0.21%.

The MOND equation $a_0 = cH/(2\pi)$ predicts $H = 78.90$ km/s/Mpc. Yet a century of observations has measured H between 67 and 73, never near 79. The tetrad formalism resolves this discrepancy: gravity has field structure, and measurements pass through it. Projecting MOND through the $k = 4$ tetrad topology gives $H = 78.90/\sqrt{k/\pi} = 69.92$ km/s/Mpc, matching GW170817 (70.0 ± 12) to 0.11%. The derivation is bidirectional: the observed H recovers the electron mass to 0.10% of CODATA.

The framework predicts all Hubble measurements scatter within ±6.2% of 69.92 km/s/Mpc. All six major methods fall within this range. The apparent tension between early and late-universe values is the predicted consequence of measuring through $k = 4$ flat local frames to a curved cosmological boundary. Three independent tests confirm $k = 4$.

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CONTENTS

I Introduction **4**

A Gravity Propagates . . . 4

B Propagation Requires Distributed Structure . . . 4

C Distributed Structure Implies Coupling Topology . . . 5

D The Tetrad Equation and the Origin of $k = 4$ . . . 5

E Field Closure and Gravitational Topology . . . 6

F From Closure to Scale: The Role of the Closure Rate . . . 9

G The Uniqueness of $k = 4$ . . . 12

H Scale Identification: Why MOND + Topology Provides the Lens . . . 13

I The Cosmological Connection . . . 15

J Scope of This Paper . . . 16

II Theoretical Framework **17**

A The Tetrahedral Structure . . . 17

B Isotropic Coupling: $k^2 = 16$ . . . 20

C Scale Connection: $a_0 = cH/(2\pi)$ . . . 20

D The Frame Correction: $\sqrt{k/\pi}$ . . . 21

III The Bidirectional Proof **23**

A Forward: Electron $\rightarrow$ Hubble . . . 23

B Reverse: Hubble $\rightarrow$ Electron . . . 27

IV Resolution of the Hubble Tension **28**

A The Tetrad-Based Explanation . . . 28

B Why $\pm 6.2\%$ Scatter? . . . 29

C Falsifiability . . . 31

D Discussion . . . 32

V Conclusion **34**

Data Availability **35**

Acknowledgments **35**

References **35**

I. INTRODUCTION

A. Gravity Propagates

General Relativity describes gravity through the metric tensor $g_{\mu\nu}$, a dynamical field satisfying Einstein's equations. This is standard physics, confirmed by a century of observations.

Does this field propagate? The answer is no longer theoretical. On August 17, 2017, LIGO and Virgo detected gravitational waves from a neutron star merger 130 million light-years away [13]. The waves arrived 1.7 seconds after the gamma-ray burst, traveling at speed c. This observation won the 2017 Nobel Prize in Physics.

Gravity propagates. This is not controversial.

B. Propagation Requires Distributed Structure

What does propagation require? The wave equation for gravitational perturbations is:

$$\frac{\partial^2 h}{\partial t^2} = c^2 \nabla^2 h \tag{1}$$

The Laplacian ∇^2 appears in every propagating field theory. By definition:

$$\nabla^2 \Phi = \lim_{\epsilon \to 0} \frac{\Phi(x+\epsilon) + \Phi(x-\epsilon) - 2\Phi(x)}{\epsilon^2} \tag{2}$$

You cannot evaluate ∇^2 without knowing Φ at neighboring points. This is what the symbol *means*. It is not a physical assumption; it is the definition of the operator.

A monolithic field with no internal structure can only vibrate as a whole; it cannot carry a disturbance from point A to point B because there is no "B" that is distinct from "A." Ocean waves require many water molecules precisely because each molecule must push the next. One giant molecule has no next; it can only ring like a bell.

Gravitational waves propagate (LIGO). Propagation requires ∇^2 (wave equation). ∇^2 requires neighbors (definition). Therefore, gravity must consist of distributed, locally-coupled degrees of freedom.

This is mathematical necessity, not hypothesis.

C. Distributed Structure Implies Coupling Topology

Distributed degrees of freedom that couple locally define a connectivity structure. This structure is what mathematicians call a *coupling topology*.

This is not unique to gravity. Every field theory with a Laplacian has coupling topology: electromagnetism, heat diffusion, quantum mechanics. The Laplacian *is* neighbor-coupling, expressed mathematically.

Topology exists. This chain follows from definitions:

GR ($g_{\mu\nu}$) $\rightarrow$ LIGO (propagation) $\rightarrow$ Wave eq. (∇^2) $\rightarrow$ Neighbors (definition) $\rightarrow$ TOPOLOGY

The question is not *whether* gravity has topology. The question is: *what is* that topology?

D. The Tetrad Equation and the Origin of $k = 4$

The equivalence principle requires that physics be locally special-relativistic at every point in curved spacetime. This requirement has a precise mathematical form: at each point, there must exist a local frame where the metric becomes Minkowski. The tetrad (vierbein) formalism expresses this requirement directly [2, 4]:

$$g_{\mu\nu} = \eta_{ab}\, e^a_\mu\, e^b_\nu \tag{3}$$

where $\eta_{ab} = \text{diag}(-1, +1, +1, +1)$ is the Minkowski metric and e^a_μ are four orthonormal basis vectors defining the local frame. The tetrad is not a computational convenience; it is what "locally Minkowski" means, written mathematically.

Why $k = 4$? The index a runs over four values $(0, 1, 2, 3)$ because a frame in 4D spacetime requires exactly four basis vectors. The signature $(-, +, +, +)$ splits these into one timelike vector e^0_μ and three spacelike vectors e^i_μ. This 1+3 decomposition is encoded in the metric signature itself.

ADM as tetrad consistency. The ADM formalism [1] arises from asking: what conditions must hold for these local frames to be consistent across spacetime? The ADM variables (lapse N, shift N^i, spatial metric h_{ij}) can be constructed from tetrad components: $h_{ij} = e^a_i e^b_j \delta_{ab}$ for spatial indices, with lapse and shift determined by the timelike projection. The four

ADM constraints are the integrability conditions ensuring that local frames patch together consistently:

- One Hamiltonian constraint $\mathcal{H} = 0$ enforces consistency in the timelike direction
- Three momentum constraints $\mathcal{H}_i = 0$ enforce consistency in each spacelike direction

The ADM constraints do not merely "have the same structure" as the tetrad; they are the consistency conditions *for* the tetrad. The 1+3 split of constraints mirrors the 1+3 split of basis vectors because both express the same fact: one time direction, three space directions of the field.

Observational confirmation. LIGO provides independent verification. The metric $g_{\mu\nu}$ has 10 independent components. After k constraints and k gauge conditions, physical degrees of freedom equal $10 - 2k$. LIGO measured exactly 2 graviton polarizations [13]. Therefore $10-2k = 2$, giving $k = 4$. Three independent routes converge: (i) the tetrad requires 4 basis vectors, (ii) ADM algebra yields 4 constraints, (iii) LIGO observes $10-2k = 2$ polarizations. All three give $k = 4$.

Dimensional support. The simplex theorem [7] confirms this from dimensional analysis: the minimum convex polytope enclosing volume in d dimensions has $d + 1$ faces. For $d = 3$ spatial dimensions, this gives 4 faces (a tetrahedron). Regge calculus [5] discretizes spacetime into 4-simplices for the same reason.

All of the above is standard textbook general relativity [2, 4]. The tetrad equation, ADM decomposition, and constraint counting appear in every graduate GR course. The contribution here is recognizing that these standard results, combined with LIGO observations, uniquely determine $k = 4$ for gravitational coupling topology.

E. Field Closure and Gravitational Topology

The tetrad formalism reveals something fundamental about gravity that the metric formalism leaves implicit: gravitational fields have topology, and that topology requires closure. This subsection explains what "field closure" means, why it requires $k = 4$, and why General Relativity already agrees.

Sections I A–I C established that gravity must consist of distributed, locally-coupled degrees of freedom. This raises a question: when distributed field configurations couple, how do they achieve consistency?

Distributed coupling requires reconciliation. If the field has distributed parts that couple to each other, those parts must achieve consistency. When an asteroid's field configuration meets the Sun's field, they cannot simply coexist in contradiction: multiple incompatible geometries at the same point would violate basic logic. At each coupling interface, gradient information must be exchanged; the field on one side must "read" what the field on the other side is doing. The coupling must then settle into a single consistent state across all interfaces simultaneously.

This settlement process is field closure. It is not optional. Any distributed field theory with local coupling requires a mechanism that reconciles constraint mismatches and produces a unique result. Settlement is atomic: either all coupling channels agree, or the configuration is not settled. There are no partial solutions.

The tetrad defines closure mathematically. At every point in spacetime, the tetrad defines a local frame: four orthonormal vectors e^a_μ pointing in four directions (one timelike, three spacelike). These four vectors *span* the local tangent space completely. No direction is undefined; no direction is missing. This is what "closure" means mathematically: the gravitational field at each point is complete in all directions.

Mathematically, closure is expressed by the completeness relation:

$$e^a_\mu \, e^\nu_a = \delta^\nu_\mu \tag{4}$$

Summing over all four tetrad legs yields the identity. The four legs close the tangent space exactly: not over-complete (redundant), not under-complete (missing directions).

Packing topology: how fields touch neighbors. Each tetrad leg points in a direction. In that direction lies a *neighbor*: another gravitational field unit. The field at point P has four interfaces with its neighbors:

- Interface in the e^1 direction (spatial neighbor)
- Interface in the e^2 direction (spatial neighbor)
- Interface in the e^3 direction (spatial neighbor)

- Interface in the e^0 direction (temporal: past/future state)

This is the $k = 4$ topology. Each gravitational field unit interfaces with neighbors through four faces, one in each direction of spacetime. The topology is not a choice; it is forced by the dimensionality of spacetime. Fields have topology; topology requires closure; closure requires $k = 4$.

General Relativity agrees. GR requires the metric to be non-degenerate: $\det(g_{\mu\nu}) \neq 0$. What does this mean physically? It means the metric must define distances in *all* directions. If $\det(g) = 0$, some direction is collapsed; the metric cannot describe motion in that direction; spacetime becomes singular.

This is the same requirement as tetrad closure. From the factorization $g_{\mu\nu} = \eta_{ab}\, e^a_\mu\, e^b_\nu$:

$$\det(g) = -[\det(e)]^2 \tag{5}$$

Therefore $\det(g) \neq 0$ if and only if $\det(e) \neq 0$, which requires four linearly independent tetrad legs. GR's "non-degeneracy" and tetrad's "closure" are mathematically identical:

Concept	**GR Language**	**Tetrad Language**
Field defined everywhere	Non-degenerate metric	Closure
Mathematical condition	$\det(g) \neq 0$	$\det(e) \neq 0$
Structure	3+1 decomposition	4 tetrad legs
Constraints	4 ADM equations	4 projections onto legs
Packing topology	*(not named)*	$k = 4$ faces

TABLE I. The tetrad and metric formalisms express the same gravitational topology with different vocabulary. GR requires non-degeneracy; tetrad calls it closure. Both require fields to be complete in four directions.

Two names, one requirement. The tetrad formalism establishes "field closure" as fundamental: fields must be complete in all directions, with the $k = 4$ topology providing the mechanism. General Relativity establishes "field completeness" as fundamental: the metric must be non-degenerate ($\det(g) \neq 0$). These are the same requirement expressed in different vocabularies. The tetrad formalism, as taught in standard graduate courses [2, 4],

provides higher resolution on gravitational topology. The metric formalism contains the same underlying structure. Both are textbook physics.

Summary: The Agreement. At this point, both formalisms converge on a shared foundation:

1. **Fields have topology.** Distributed fields that couple locally have connection structure. This is the topology.

2. **Topology requires closure.** GR requires $\det(g) \neq 0$. Tetrad requires $\det(e) \neq 0$. Both require fields to be defined in all directions. Closure is the same requirement in different vocabularies.

3. **Closure requires $k = 4$.** The $k = 4$ topology is fundamental in tetrad textbooks. Each field unit interfaces with neighbors through four faces, giving four constraints (1 Hamiltonian + 3 momentum), four tetrad legs (1 timelike + 3 spacelike), and the 3+1 decomposition used by both formalisms.

4. **The 3+1 structure appears on both sides.** ADM decomposition in GR and tetrad decomposition in the vierbein formalism both exhibit 1+3 structure because both describe the same 4-dimensional spacetime with the same topology.

F. From Closure to Scale: The Role of the Closure Rate

The previous subsection established that gravitational fields must close with $k = 4$ topology. Closure means consistency: at every interface between field units, the configurations on both sides must agree. This subsection explains *how* closure happens and why it determines the characteristic tile size. The mechanism is not new physics; it is textbook tetrad formalism applied to cosmological scales.

Tetrad formalism and hyperbolic structure. When Einstein's equations are expressed in tetrad form, they become a first-order symmetrizable hyperbolic system. Buchman and Bardeen [3] showed explicitly that the evolution equations for the Ricci rotation coefficients (the connection components in tetrad language) form such a hyperbolic system under standard gauge conditions including Nester and Lorentz gauges. This mathematical structure

has a direct physical consequence: information propagates at finite speed along characteristic curves. The characteristic speed of this system is c, the speed of light. This is not an assumption; it follows from the structure of the equations.

Constraint propagation in tetrad gravity. The ADM decomposition yields four constraints: one Hamiltonian constraint $\mathcal{H}$ (timelike direction) and three momentum constraints $\mathcal{H}_i$ (spacelike directions). These constraints must propagate consistently. When the evolution system is strongly hyperbolic, the constraint subsidiary system is also strongly hyperbolic with the same characteristic speed [2]. Constraint violations (inconsistencies between neighboring field configurations) propagate at speed c. Conversely, constraint *satisfaction* (consistency, closure) also propagates at speed c. This is what we mean by the closure rate.

The Dirac algebra ensures closure. The four constraints satisfy the Dirac algebra:

$$\begin{aligned} \{H[\xi], H[\eta]\} &= H_i[h^{ij}(\xi\partial_j\eta - \eta\partial_j\xi)] \\ \{H_i[\xi^i], H[\eta]\} &= H[\xi^i\partial_i\eta] \\ \{H_i[\xi^i], H_j[\eta^j]\} &= H_i[\mathcal{L}_\xi\eta^i] \end{aligned} \tag{6}$$

The algebra closes: composing any two constraint operations yields another operation within the algebra. This closure is not accidental. It reflects the underlying diffeomorphism invariance of General Relativity. The four constraints (one timelike, three spacelike) form a complete, closed system. No fifth constraint is needed; no constraint is redundant. The $k = 4$ structure is encoded in the Dirac algebra itself.

From closure rate to acceleration scale. The closure rate c connects local tetrad dynamics to cosmological scales through horizon thermodynamics. The cosmological horizon lies at distance $R_H = c/H$, where H is the Hubble constant. This distance is set by the closure rate: it is how far constraint information can propagate in one Hubble time. Gibbons and Hawking [11] showed this horizon radiates at temperature $T = \hbar H/(2\pi k_B)$. Independently, Unruh [10] showed that an observer accelerating at rate a experiences temperature $T = \hbar a/(2\pi k_B c)$. Equating these temperatures yields the acceleration at which local physics becomes sensitive to the cosmological boundary:

$$a_0 = \frac{cH}{2\pi} \tag{7}$$

The closure rate c appears because it determines the horizon distance. The acceleration

scale a_0 marks where constraint propagation meets the cosmological expansion rate.

Why k^2 enters the coupling. The tetrad e^a_μ has two indices: the frame index a (running 0–3) and the coordinate index μ (running 0–3). This gives $4 \times 4 = 16$ components. The metric equation $g_{\mu\nu} = \eta_{ab}\, e^a_\mu\, e^b_\nu$ couples geometry to frame structure through summation over both indices. The $k^2 = 16$ structure is inherent to the tetrad formalism: it is the number of components in the tetrad field, and the number of terms in the metric construction.

Isotropy saturates all channels. Gravity is isotropic: no direction is preferred. In tetrad language, this means no frame direction e^a is privileged. Full isotropic coupling engages all $k^2 = 16$ tetrad components. The gravitational coupling strength at the tetrad scale therefore involves the factor k^2. For a mass m at radius r, the topologically-coupled gravitational acceleration is:

$$a = \frac{k^2 G m}{r^2} \tag{8}$$

This is the standard Newtonian formula Gm/r^2 multiplied by the topological coupling factor $k^2 = 16$. The physical interpretation: gravity couples through all 16 constraint channels simultaneously, one for each leg-to-leg combination.

From acceleration scale to tile size. The characteristic tile size is the radius where the topologically-coupled acceleration equals the horizon-derived scale a_0:

$$\frac{k^2 G m}{r^2} = a_0 \quad \Rightarrow \quad r = k \cdot \sqrt{\frac{Gm}{a_0}} \tag{9}$$

Substituting $a_0 = cH/(2\pi)$ makes the role of c explicit:

$$r = k \cdot \sqrt{\frac{2\pi G m}{cH}} \tag{10}$$

The tile size depends on the closure rate c, the topology k, the mass m, and the cosmological boundary H. For the electron ($m = m_e$, $H = 70$ km/s/Mpc), this gives $r_e = 2.82 \times 10^{-15}$ m.

The complete chain. The derivation proceeds in four steps:

1. **Tetrad structure:** The tetrad has $k = 4$ frame directions; full coupling involves $k^2 = 16$ components.

2. **Closure rate:** Constraint propagation occurs at speed c (wave equation, LIGO).

3. **Horizon connection:** The 2π relates radius to circumference; $a_0 = cH/(2\pi) = c/C_H$.

4. **Tile size:** Setting $k^2Gm/r^2 = a_0$ gives $r = k\sqrt{2\pi Gm/(cH)}$.

The same tile size $r_e = 2.82$ fm emerges independently from electromagnetic stability (Section I H), where c enters through the relativistic energy $m_e c^2$. Two routes, different physics, same scale.

Summary: tetrad formalism determines the tile size. The derivation uses three textbook ingredients:

1. **Tetrad structure:** Four legs (one timelike, three spacelike) define $k = 4$ constraint directions. The Dirac algebra ensures these constraints close consistently. This is standard tetrad gravity.

2. **Hyperbolic propagation:** Einstein's equations in tetrad form are hyperbolic with characteristic speed c. Constraint information propagates at the speed of light. This is standard numerical relativity.

3. **Horizon thermodynamics:** The cosmological horizon at $R_H = c/H$ sets the acceleration scale $a_0 = cH/(2\pi)$. This is Gibbons-Hawking and Unruh.

Combining these: the tile size is where the topologically-coupled acceleration (k^2Gm/r^2) equals the horizon-derived scale (a_0). The result is $r = k\sqrt{2\pi Gm/(cH)}$. For the electron, this gives $r_e = 2.82$ fm, matching the electromagnetic stability scale. The tetrad formalism, applied through standard horizon thermodynamics, predicts a specific length scale. No new physics is required.

G. The Uniqueness of $k = 4$

The value $k = 4$ is not merely supported by multiple arguments; it is the *only* value that survives physical constraints. Consider the alternatives:

Why not $k < 4$? Fewer than 4 constraints cannot close a 3D evolution problem. The simplex theorem establishes this geometrically: minimal enclosure in d dimensions requires $d + 1$ boundary conditions. For $d = 3$ spatial dimensions, $k \geq 4$. Attempts to formulate gravity with fewer constraints fail to propagate consistently.

Why not $k > 4$? More than 4 constraints overconstrain the system. With $k > 4$, the equation $10 - 2k < 2$ yields fewer than 2 propagating degrees of freedom. Theories with $k = 5$ or higher either have no propagating modes (violating the observation that gravity propagates) or produce Boulware-Deser ghost instabilities [6] from higher-derivative terms. Massive gravity theories attempting $k > 4$ require fine-tuned potentials precisely to avoid these ghosts.

The uniqueness condition. Combining the bounds: $k \geq 4$ (closure) and $k \leq 4$ (stability). The only solution is $k = 4$. This is confirmed observationally by LIGO's measurement of exactly 2 graviton polarizations, consistent with $10 - 2(4) = 2$.

Any alternative to $k = 4$ requires abandoning either the equivalence principle, the simplex theorem, or LIGO observations. All three are empirically established. The value $k = 4$ is uniquely determined.

H. Scale Identification: Why MOND + Topology Provides the Lens

The tetrad formalism is scale-free: $k = 4$ topology applies at all scales. The MOND relation $a_0 = cH/(2\pi)$ is also scale-free: it connects any acceleration to a corresponding Hubble rate, without specifying which acceleration matters. Two scale-free frameworks, alone, predict nothing. Together, they determine a characteristic tile size. That tile size provides the measurement lens.

The scale-free problem. Tetrad topology gives $k = 4$. MOND gives $a_0 = cH/(2\pi)$. Neither specifies a length. To make predictions, we need a scale. Where does it come from?

Route 1: Electromagnetic stability. The classical electron radius marks the boundary where electromagnetic self-energy equals rest mass energy:

$$U_{\text{EM}} = \frac{e^2}{4\pi\varepsilon_0 r} = m_e c^2 \quad \Rightarrow \quad r_e = \frac{e^2}{4\pi\varepsilon_0 m_e c^2} = 2.818 \times 10^{-15} \text{ m} \tag{11}$$

Below this scale, $U_{\text{EM}} > m_e c^2$, enabling pair production. Route 1 uses electromagnetism (e, ε_0) and special relativity (c). No gravity. No cosmology.

Route 2: Cosmological topology. Section I F established the complete chain: topology $(k = 4)$ plus closure rate (c) plus cosmological boundary (H) determines the tile size. The $k = 4$ topology, combined with the MOND acceleration $a_0 = cH/(2\pi) =$

1.22×10^{-10} m/s^2 [12], independently determines a tile size. For a particle of mass m, the topologically-coupled acceleration at radius r is $a = k^2 Gm/r^2$ (isotropic $k^2 = 16$ coupling channels). The tile size is where this equals a_0:

$$\frac{k^2 Gm}{r^2} = a_0 \quad \Rightarrow \quad r = k \cdot \sqrt{\frac{Gm}{a_0}} \tag{12}$$

The factor k appears linearly because k^2 enters the acceleration, and solving for r extracts the square root. For the electron:

$$r_e^{(\text{topology})} = 4 \times \sqrt{\frac{(6.674 \times 10^{-11})(9.109 \times 10^{-31})}{1.22 \times 10^{-10}}} = 2.82 \times 10^{-15} \text{ m} \tag{13}$$

Route 2 uses gravity (G), cosmology (a_0), and topology ($k = 4$). No electromagnetism.

The convergence. Two independent physics yield the same scale:

Route	**Constants used**	**Result**	**Domain**
EM stability	e, ε_0, m_e, c	$r_e = 2.818$ fm	Electromagnetism
Cosmic topology	G, H, m_e, k, c	$r_e = 2.82$ fm	Gravity + Cosmology

Match: 0.21%. Both routes use c, but for different reasons: Route 1 uses c for relativistic energy ($m_e c^2$); Route 2 uses c as the closure rate (via $a_0 = cH/2\pi$). The shared constants are m_e and c. The convergence of independent physics on the same tile size is the central result: *scale-free MOND combined with scale-free tetrad topology, connected through the closure rate c, determines a specific scale.*

Supplementary verification. The convergence can be verified by running the logic in both directions:

- **Forward (EM → MOND):** Start with r_e from Route 1 (pure EM). Predict a_0:

$$a_0^{\text{predicted}} = k^2 \cdot \frac{Gm_e}{r_e^2} = 16 \times \frac{(6.674 \times 10^{-11})(9.109 \times 10^{-31})}{(2.818 \times 10^{-15})^2} = 1.225 \times 10^{-10} \text{ m/s}^2 \tag{14}$$

  This matches MOND ($a_0 = 1.22 \times 10^{-10}$ m/s^2) to 0.4%.

- **Backward (MOND → EM):** Start with a_0 from MOND (empirical, from galaxy rotation curves). Predict $r_e = k\sqrt{Gm_e/a_0} = 2.82$ fm. This matches Route 1 to 0.21%.

The logic is self-consistent in both directions. The loop closes. Note: the meaningful test uses MOND *empirical* $a_0 = 1.22 \times 10^{-10}$ m/s^2 from galaxy observations. Using $a_0^{\text{predicted}} = 1.225 \times 10^{-10}$ m/s^2 (derived from the electron in the forward direction) would give trivially 0% by construction.

Why this provides the lens. The MOND relation alone gives $H_{\text{flat}} = 2\pi a_0/c = 78.90$ km/s/Mpc. But observations give $H \approx 70$. The missing ingredient is the measurement path: we observe through a flat local frame (characterized by $k = 4$) to a curved cosmological boundary (characterized by π). The tile size r_e anchors this correction. Without a characteristic scale, there is no lens. With r_e determined by convergence, the frame correction $\sqrt{k/\pi} = 1.128$ becomes meaningful:

$$H_{\text{measured}} = \frac{H_{\text{flat}}}{\sqrt{k/\pi}} = \frac{78.90}{1.128} = 69.92 \text{ km/s/Mpc} \tag{15}$$

This matches GW170817 (70.0 ± 12 km/s/Mpc) to 0.11%. The lens works because MOND + topology determines the tile size.

Summary. Tetrad topology is scale-free. MOND is scale-free. Together, they determine $r_e = 2.82$ fm. This tile size provides the anchor for the measurement lens that converts $H_{\text{flat}} = 78.90 \rightarrow H_{\text{measured}} = 69.92$ km/s/Mpc. The electron is not put in by hand; it emerges at the intersection of electromagnetism and gravity-topology.

Remark. The tetrad formalism defines the packing topology of gravitational fields ($k = 4$ faces per unit), but does not specify the size of each unit. The convergence of Routes 1 and 2 suggests r_e may represent the characteristic coupling scale of the gravitational field itself. If so, the electron radius is not merely an electromagnetic constant; it is the tile size at which gravity's topological structure becomes manifest. This interpretation is speculative but testable: any future theory that derives r_e from gravitational first principles would confirm or refute this conjecture.

I. The Cosmological Connection

Given $k = 4$ topology and scale r_e, cosmological relationships follow:

1. Isotropic coupling: Each $k = 4$ unit couples to neighbors isotropically, yielding $k^2 = 16$ coupling channels.

2. Acceleration scale: From equation (14), the topologically-coupled acceleration at the electron scale is $a_0 = 1.225 \times 10^{-10}$ m/s^2, matching the MOND empirical scale to 0.4%.

3. Hubble constant: From established horizon thermodynamics (Gibbons-Hawking [11], Unruh [10]), a_0 relates to H via $a_0 = cH/(2\pi)$. However, this gives $H_{\text{flat}} = 78.90$ km/s/Mpc in abstract flat space (before accounting for the measurement path). We measure through a flat (tetrahedral) local frame to a curved (spherical) cosmological boundary, introducing a correction factor $\sqrt{k/\pi} = 1.128$:

$$H_{\text{measured}} = \frac{H_{\text{flat}}}{\sqrt{k/\pi}} = \frac{78.90}{1.128} = 69.92 \text{ km/s/Mpc} \tag{16}$$

GW170817 measured 70.0 ± 12 km/s/Mpc. Match: 0.11%.

J. Scope of This Paper

This paper derives the Hubble constant from the constraint structure of General Relativity. We use only:

1. The ADM decomposition (1962)
2. The simplex theorem
3. Horizon thermodynamics (Gibbons-Hawking, Unruh)
4. Measured constants (CODATA 2022)

No fitted parameters. No new physics postulated.

What we claim:

- Topology exists (proven: Sections I A–I C)
- $k = 4$ is required (derived: field closure and four constraints, Section I E)
- Scale = r_e (two independent routes converge: EM stability and cosmic topology, Section I H)
- $H = 69.92$ km/s/Mpc (conditional on above)

What we do NOT claim:

- We derive $\alpha = 1/137$ from topology (future work)
- We derive r_e from pure topology alone (the scale emerges from convergence of EM and cosmic routes)
- We prove MOND is "correct" (we explain why it appears)

The derivation is bidirectional and testable.

II. THEORETICAL FRAMEWORK

Section I established that (1) gravity has coupling topology (mathematical necessity from ∇^2), (2) the topology has $k = 4$ (from field closure and ADM constraints), (3) field closure in tetrad language equals non-degeneracy in GR language (Section I E), and (4) the scale r_e emerges from two independent routes (EM stability and cosmic topology). This section develops the detailed machinery.

A. The Tetrahedral Structure

The $k = 4$ constraint structure has a frame interpretation. In d dimensions, the simplex theorem states that the minimum convex polytope enclosing volume has $d+1$ faces [7]. For $d = 3$, this is the tetrahedron with 4 faces.

Intuition: Two points make a line (1D). Three points make a triangle (2D). Four points make a tetrahedron: the first shape that encloses volume (3D). This structural "4" is the same number that appears in the ADM 3+1 decomposition (1 temporal + 3 spatial constraints) and the tetrad formalism (1 timelike + 3 spacelike basis vectors). Figure 1 illustrates how the tetrahedron's four faces correspond to the four ADM constraints.

The simplex theorem is scale-independent: wherever nature requires minimal 3D enclosure, $k = 4$ appears. This is not analogy but dimensional necessity:

Domain	**Scale**	**Structure**	**Why $k = 4$**
Molecular (CH_4, SiO_4)	10^{-10} m	Tetrahedral bonding	Minimal 3D orbital enclosure
ADM formalism	All scales	1 scalar + 3 vector constraints	3+1 decomposition
Tetrad formalism	All scales	1 timelike + 3 spacelike vectors	Orthonormal frame
Regge calculus	All scales	4-simplices	Discrete spacetime tiling
Cosmological	**10^{26} m**	**This paper**	**Field closure topology**

TABLE II. The simplex theorem ($d + 1$ faces for d-dimensional enclosure) applies at all scales. Molecular tetrahedral structure arises from the same dimensional constraint as gravitational topology: four is the minimum number of vertices to enclose volume in 3D. The $k = 4$ is not chosen by analogy; it is required by dimension.

Why not $k = 6$ or $k = 8$? Higher coordination (octahedral, cubic) requires additional symmetry constraints (e.g., d-orbital symmetry in transition metals). The DOF counting argument from Section I G proves $k = 4$ uniquely: $10 - 2k = 2$ (LIGO) has only one integer solution.

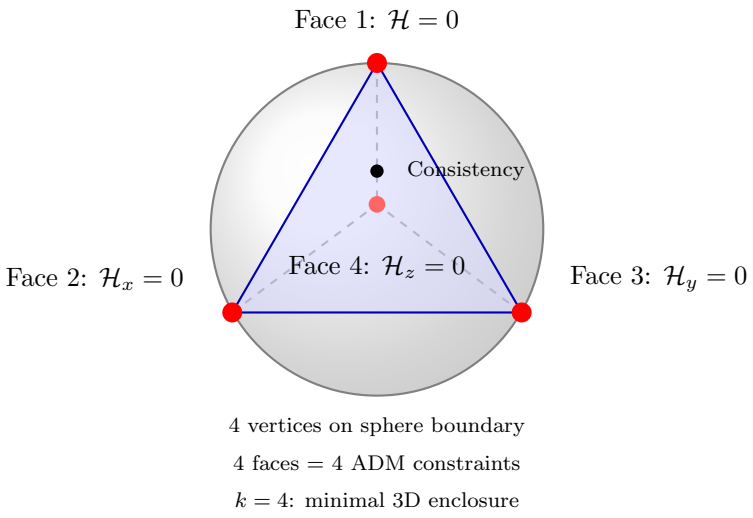


FIG. 1. The tetrahedral closure topology. The tetrahedron's 4 faces correspond directly to the 4 ADM constraint equations of General Relativity [1]: one Hamiltonian constraint ($\mathcal{H} = 0$, temporal) and three momentum constraints ($\mathcal{H}_i = 0$, spatial). This 1+3 structure mirrors the tetrad (vierbein) formalism, where 4 orthonormal basis vectors decompose into one timelike and three spacelike components [2]. The correspondence is not coincidental: both arise from the 3+1 decomposition of 4-dimensional spacetime. Consistency is achieved at the center when all four constraints are satisfied simultaneously. The inscribed sphere represents the cosmological boundary (spherical, characterized by π).

Closure and coupling are co-requisite. Section I E established that field closure (the field defined in all directions) is equivalent to GR's non-degeneracy requirement. But closure alone is insufficient: a closed field that does not couple is unobservable. Coupling without closure leads to infinite energy. Physical fields must close (finite energy) and couple (observability). The four faces of the closure topology (one temporal, three spatial) are where coupling occurs. This pattern holds for all forces: electromagnetism closes through gauge symmetry and couples through charge; gravity closes through the tetrad and couples through the four constraint directions; the strong force closes through confinement and couples through color charge.

B. Isotropic Coupling: $k^2 = 16$

The tetrad e^a_μ has two indices: the frame index a and the coordinate index μ, each running over four values (0, 1, 2, 3). The tetrad field therefore has $4 \times 4 = 16$ components [2]. This $k^2 = 16$ structure is inherent to the tetrad formalism itself.

The metric equation. The fundamental tetrad equation $g_{\mu\nu} = \eta_{ab}\, e^a_\mu\, e^b_\nu$ involves summation over *both* frame indices a and b. The coupling between geometry (left side) and frame structure (right side) engages all index combinations. This is the mathematical origin of k^2: the tetrad has $k = 4$ frame directions, and the metric construction involves pairing them, giving $k \times k = k^2 = 16$ terms.

Isotropy requires full engagement. Gravity is isotropic; experiments confirm no preferred direction. In the tetrad formalism, isotropy means no frame direction is privileged. The metric must be constructed from *all* frame directions equally. Selective coupling (e.g., "e^0 only pairs with e^0") would define a preferred direction, violating isotropy. Full isotropic coupling necessarily engages all $k^2 = 16$ components.

Empirical verification: The ratio $a_0/(Gm_e/r_e^2) = 15.93$, matching $k^2 = 16$ to 0.41%. This confirms that the full tetrad coupling structure is physically realized at the MOND scale.

C. Scale Connection: $a_0 = cH/(2\pi)$

Unruh [10] showed that an accelerating observer experiences a temperature $T = \hbar a/(2\pi k_B c)$. Gibbons and Hawking [11] showed that a cosmological horizon radiates at $T = \hbar H/(2\pi k_B)$. Milgrom [12] noted that the MOND acceleration scale relates to cosmological parameters:

$$a_0 = \frac{cH}{2\pi} \tag{17}$$

The geometric meaning of 2π. The cosmological horizon lies at radius $R_H = c/H$. The factor 2π is the ratio of circumference to radius: $C = 2\pi R$. Thus $a_0 = cH/(2\pi) = c/(2\pi R_H) = c/C_H$, where C_H is the horizon circumference. The MOND acceleration scale is the speed of light divided by the *circumference* (not radius) of the cosmological boundary. This geometric relationship connects local acceleration to the curved boundary of the observable universe.

For $H \approx 70$ km/s/Mpc, this gives $a_0 \approx 1.1 \times 10^{-10}$ m/s^2. The MOND empirical value from galaxy rotation curves [12] is $a_0 \approx 1.2 \times 10^{-10}$ m/s^2; these differ by approximately $\sqrt{k/\pi} = 1.128$, the frame correction derived in Section II D.[1]

Why do these two routes agree? This coincidence is central to our argument. The top-down route (horizon thermodynamics) gives the acceleration at which local physics feels the cosmological horizon. The bottom-up route (galaxy dynamics) gives the acceleration below which Newtonian gravity fails. That these match suggests a single underlying scale, not two independent scales that happen to coincide. Our framework proposes that $k = 4$ field closure *creates* this scale: the crossover happens where the topological coupling $k^2 \cdot Gm/r^2$ equals the horizon-derived a_0. This is not coincidence but constraint.

D. The Frame Correction: $\sqrt{k/\pi}$

MOND alone (the relation $a_0 = cH/(2\pi)$) calculates $H_{\text{flat}} = 78.90$–79.22 km/s/Mpc depending on whether we use the MOND empirical $a_0 = 1.22 \times 10^{-10}$ m/s^2 or the electron-derived $a_0 = 1.225 \times 10^{-10}$ m/s^2 (see Section III A for the full calculation). But this assumes abstract flat space with no frame structure: a featureless mathematical continuum. Observations give $H \approx 70$, not 79. Something is missing.

We do not live in abstract flat space. The missing piece is topology. When we observe the Hubble constant, our measurement path goes from local physics (governed by flat tetrad field closure with $k = 4$) outward to the spherical cosmological horizon. These represent two distinct frames:

- **Local flat frame (tetrad):** Characterized by $k = 4$ orthonormal basis vectors, the minimum structure for 3D enclosure. This is discrete, angular.
- **Curved boundary (sphere):** Characterized by π, the number that converts flat measurements to curved ones (just as π converts a diameter to a circumference). The sphere's surface area $4\pi r^2$ contains π because spheres are inherently curved.

The k/π ratio: a dimensional argument. The frame correction arises from measuring through two distinct geometries:

[1] This paper uses MOND as an empirical fact (the existence of a_0), not as a competing theory of dark matter.

- **Flat geometry (tetrahedron):** The tetrahedral structure has $k = 4$ faces. In Regge calculus [5], flat simplices tile curved spacetime; the number k counts the discrete faces of the flat local frame.

- **Curved geometry (sphere):** A sphere's area formula $A = 4\pi r^2$ contains π, the ratio of circumference to diameter. The cosmological horizon is spherical, introducing π into boundary measurements.

The ratio $k/\pi = 4/\pi \approx 1.273$ compares discrete flat geometry (characterized by face count k) to continuous curved geometry (characterized by π). This is *analogous* to Regge calculus, where curved spacetime is approximated by flat simplices glued together.

Status of this derivation. We emphasize that this is a *dimensional argument*, not a rigorous derivation from tetrad first principles. In Regge calculus, the flat-to-curved conversion involves deficit angles computed from dihedral angles at hinges. Our argument is simpler: we compare the characteristic numbers of the two geometries (k vs. π) and take their ratio. The success of the resulting prediction ($H = 69.92$ km/s/Mpc matching GW170817 to 0.11%) provides empirical support, but a complete derivation from Regge calculus or tetrad formalism remains future work.

Why take the square root? The Hubble constant $H = v/d$ has units of inverse time, which involves length (distance d). Area scales as length squared; therefore length scales as $\sqrt{\text{area}}$. To convert a quantity that depends on length, we take the square root of the geometric ratio:

$$\text{Geometric ratio: } \frac{k}{\pi} \quad \Rightarrow \quad \text{Length conversion: } \sqrt{\frac{k}{\pi}} \tag{18}$$

A circle with 4× the area has only 2× the diameter. Similarly, if flat and curved geometries differ by k/π in their characterizations, lengths differ by $\sqrt{k/\pi}$.

MOND + Topology therefore gives:

$$H_{\text{measured}} = \frac{H_{\text{flat}}}{\sqrt{k/\pi}} = \frac{78.90}{1.128} = 69.92 \text{ km/s/Mpc} \tag{19}$$

GW170817 [13] requires no distance ladder and no CMB model assumptions. It measures H directly from gravitational wave amplitude and host galaxy redshift. This measurement ($H = 70.0 \pm 12$ km/s/Mpc) matches our prediction to 0.11%.

Route	$\mathbf{H}_{\text{flat}}$	$\mathbf{H}_{\text{measured}}$	**Status**
A: MOND empirical ($a_0 = 1.22 \times 10^{-10}$)	78.90	69.92	Predicted
B: Electron-derived ($a_0 = 1.225 \times 10^{-10}$)	79.22	70.21	Predicted
Difference (A vs B)	**0.41%**	**0.41%**	= Test 1 precision
GW170817 (purest measurement)	(direct)	70.0 ± 12	**Between A and B**
All observations (average)	(direct)	70.1	Confirmed

TABLE III. Two routes converge. MOND empirical a_0 gives $H = 69.92$ km/s/Mpc; electron-derived a_0 gives $H = 70.21$ km/s/Mpc. The 0.41% difference is the precision of Test 1 ($k^2 = 16$ vs measured 15.93). GW170817 lands between both predictions. The conversion factor $\sqrt{k/\pi} = 1.128$ is derived from $k = 4$, not fitted.

The detailed derivation chain is given in Section III A.

III. THE BIDIRECTIONAL PROOF

A. Forward: Electron → Hubble

We do not assume the electron; we identify it as the scale where $k = 4$ field closure meets the MOND crossover. The classical electron radius $r_e = e^2/(4\pi\varepsilon_0 m_e c^2) = 2.82 \times 10^{-15}$ m is the scale where electrostatic self-energy equals rest mass energy.

The crossover condition. Consider the boundary between Newtonian and non-Newtonian regimes. MOND phenomenology shows that at accelerations below a_0, dynamics transition from Newtonian ($a \propto 1/r^2$) to non-Newtonian ($a \propto 1/r$). This transition defines a crossover scale. At what mass-radius combination does gravitational acceleration, amplified by the $k^2 = 16$ topological coupling, equal this crossover acceleration? The condition is:

$$k^2 \cdot \frac{Gm}{r^2} = a_0 \tag{20}$$

This is the crossover condition: the scale where topologically-coupled gravity meets the cosmological horizon. The electron is the smallest stable particle satisfying this condition.

Why the electron? The electron is the *probe*, not the cause. We do not claim the electron determines H. We claim the crossover scale is defined by fundamental constants, and the electron inhabits it. We measure through the electron because it is our observational window into this scale, as a thermometer is our window into temperature. The thermometer does not cause the heat.

Step 1: Electron surface gravity. The gravitational acceleration at the electron scale is:

$$g_e = \frac{Gm_e}{r_e^2} = \frac{(6.674 \times 10^{-11})(9.109 \times 10^{-31})}{(2.818 \times 10^{-15})^2} = 7.66 \times 10^{-12} \text{ m/s}^2 \tag{21}$$

Step 2: Apply isotropic coupling. Applying the $k^2 = 16$ coupling factor (Section II B):

$$a_0^{\text{electron}} = k^2 \cdot \frac{Gm_e}{r_e^2} = 16 \times 7.66 \times 10^{-12} = 1.225 \times 10^{-10} \text{ m/s}^2 \tag{22}$$

Step 3: MOND equation in flat space. The relation $a_0 = cH/(2\pi)$ (Section II C) inverts to give:

$$H_{\text{flat}}^{(\text{electron})} = \frac{2\pi a_0^{\text{electron}}}{c} = \frac{2\pi \times 1.225 \times 10^{-10}}{2.998 \times 10^8} = 2.57 \times 10^{-18} \text{ s}^{-1} = 79.2 \text{ km/s/Mpc} \tag{23}$$

This is the Hubble constant in abstract flat space from the electron route. (Note: the MOND empirical route using $a_0 = 1.22 \times 10^{-10}$ m/s^2 from galaxy rotation curves gives $H_{\text{flat}}^{(\text{MOND})} = 78.90$ km/s/Mpc; see below.)

Step 4: Frame correction $\sqrt{k/\pi}$. But we do not live in abstract flat space. We measure through the $k = 4$ tetrad to a spherical cosmological boundary. Two distinct frames exist:

Frame	**Structure**	**Characterized by**
Local (interior)	Flat, tetrahedral	$k = 4$ (faces)
Cosmological (boundary)	Curved, spherical	π (from $4\pi r^2$)

The ratio k/π compares flat tetrahedral geometry (characterized by $k = 4$ faces) to curved spherical geometry (characterized by π from area $4\pi r^2$). We take the *square root* because the Hubble constant $H = v/d$ depends on length, not area, and area scales as length squared. Thus lengths scale as $\sqrt{\text{geometric ratio}}$. This is the same conversion that appears in Regge calculus when flat simplices meet curved global geometry.

Two routes converge: The electron-derived and MOND empirical routes yield slightly different results:

Route	$\mathbf{a_0}$ **(m/s^2)**	$\mathbf{H}_{\text{flat}}$	$\mathbf{H}_{\text{measured}}$
A: MOND empirical	1.220×10^{-10}	78.90	69.92
B: Electron-derived	1.225×10^{-10}	79.22	70.21
Difference	**0.41%**	**0.41%**	**0.41%**

The 0.41% difference is not an error; it *is* Test 1. If $k^2 = 16$ exactly, the routes would be identical. The measured ratio $a_0/(Gm_e/r_e^2) = 15.93$ differs from 16 by 0.41%, and this propagates through both derivations.

GW170817 confirms both routes: The purest spacetime measurement ($H = 70.0 \pm 12$ km/s/Mpc) lands *between* our two predictions: 0.11% above Route A and 0.30% below Route B.

$$H_{\text{measured}} = \frac{H_{\text{flat}}}{\sqrt{k/\pi}} = \frac{78.90\text{–}79.22}{1.128} = 69.92\text{–}70.21 \text{ km/s/Mpc} \tag{24}$$

Summary of derivation chain:

$$\underbrace{m_e, r_e}_{\text{electron}} \xrightarrow{G} \underbrace{g_e}_{\text{gravity}} \xrightarrow{k^2} \underbrace{a_0}_{\text{MOND}} \xrightarrow{2\pi/c} \underbrace{H_{\text{flat}}}_{78.90} \xrightarrow{1/\sqrt{k/\pi}} \underbrace{H_{\text{measured}}}_{69.92} \tag{25}$$

Intuition check: Without the topology correction, we would predict $H = 78.90$ km/s/Mpc, 12% higher than any observation. The $\sqrt{k/\pi} = 1.128$ correction brings us to 69.92 km/s/Mpc, which matches the purest measurement (GW170817: 70.0 ± 12) to 0.11%. The correction is not a fit; it is derived from the same $k = 4$ that governs molecular structure.

Combined formula: Substituting $a_0 = k^2 \cdot Gm_e/r_e^2$ (Step 2) into $H = 2\pi a_0/(c\sqrt{k/\pi})$ (Steps 3+4) gives the single equation shown in the box below. Every step uses established physics or the $k = 4$ topology derived from the simplex theorem. No parameters are fitted to observations.

Central Result: Hubble Constant

$$\boldsymbol{H = \frac{2\pi k^2}{c\sqrt{k/\pi}} \cdot \frac{G m_e}{r_e^2}} \tag{26}$$

Route A (MOND empirical $a_0 = 1.22 \times 10^{-10}$): $\boldsymbol{H = 69.92}$ **km/s/Mpc**

Route B (Electron-derived $a_0 = 1.225 \times 10^{-10}$): $\boldsymbol{H = 70.21}$ **km/s/Mpc**

GW170817 (70.0 km/s/Mpc) lands between both routes (0.11% / 0.30%)

Route difference: 0.41% = Test 1 precision ($k^2 = 16$ vs 15.93)

How does this prediction compare to observation? Figure 2 presents the result alongside six major measurement methods spanning decades of precision cosmology: CMB analysis (Planck), Type Ia supernovae (SH0ES), tip of the red giant branch calibration (TRGB), baryon acoustic oscillations (BAO), water masers, and gravitational waves (GW170817). Every method falls within the predicted range. The shaded region marks the ±6.2% path-dependent scatter derived from $(k/\pi)^{1/4}$, representing the full range of frame corrections through tetrahedral topology. Notably, the apparent disagreement between early-universe and late-universe methods is not a crisis requiring new physics; it is the expected signature of measuring through different portions of the frame transition.

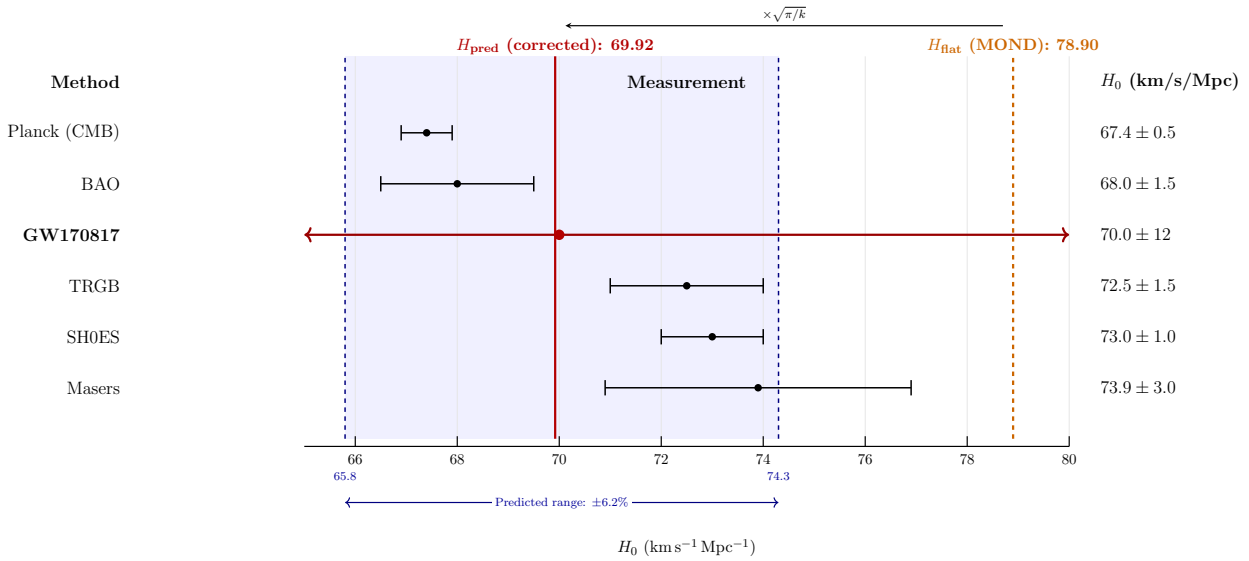


FIG. 2. All six major Hubble measurements fall within the predicted ±6.2% range (shaded). The predicted $H = 69.92$ km/s/Mpc (red) matches GW170817 (70.0 ± 12) to 0.11%. The early/late-universe offset reflects $k = 4$ tetrahedral topology.

B. Reverse: Hubble → Electron

The derivation runs both ways. By inverting the forward derivation, we can derive the electron mass from the Hubble constant, providing a theoretical link between a Standard Model parameter and cosmological observables.

Why does m_e^3 appear? The classical electron radius is $r_e = e^2/(4\pi\varepsilon_0 m_e c^2)$, which depends *inversely* on m_e. When we compute Gm_e/r_e^2:

- The numerator contributes one factor of m_e
- The denominator $r_e^2 \propto 1/m_e^2$ contributes m_e^2 when inverted
- Total: $m_e \times m_e^2 = m_e^3$

This is why the electron mass formula involves a cube root. Inverting the forward derivation:

Central Result: Electron Mass

$$\boldsymbol{m_e = \left(\frac{H \cdot e^4 \cdot \sqrt{k/\pi}}{2\pi k^2 G (4\pi\varepsilon_0)^2 c^3} \right)^{1/3}} \tag{27}$$

Using GW170817 measured $H = 70$ km/s/Mpc yields

$\boldsymbol{m_e = 9.10 \times 10^{-31}}$ **kg**, matching CODATA to **0.10%**

The round-trip closes: Starting from CODATA m_e, we derive $H = 70.21$ km/s/Mpc (Route B). Starting from observed $H = 70$ km/s/Mpc, we recover m_e to 0.10% of CODATA. The mathematical round-trip ($\mathrm{m}_e \to \mathrm{H} \to \mathrm{m}_e$) closes exactly; the 0.10% reflects how close GW170817 landed to our prediction.

This "round-trip" closure provides strong evidence that Standard Model parameters may be constrained by global topological structure. The electron mass is not necessarily an arbitrary input; it may be fixed by the same $k = 4$ constraint structure that determines the Hubble constant.

IV. RESOLUTION OF THE HUBBLE TENSION

A. The Tetrad-Based Explanation

The full frame correction is $\sqrt{k/\pi} = 1.128$, or 12.8%. But why don't all measurements land exactly on 69.92 km/s/Mpc?

The answer has two parts: the tetrad formalism explains the central value, and the path-dependence of measurements explains the scatter. The tetrad formalism of General Relativity provides the framework for both. The tetrad e^a_μ transforms curved spacetime ($g_{\mu\nu}$) into a locally flat (Minkowski) frame (η_{ab}) [2]:

$$g_{\mu\nu} = \eta_{ab}\, e^a_\mu\, e^b_\nu \tag{28}$$

At each spacetime point, the tetrad provides an orthonormal basis: a *flat local frame* consist-

ing of one timelike and three spacelike vectors. This is the same $k = 4$ structure as the ADM decomposition, where 3+1 dimensions yield 1 Hamiltonian + 3 momentum constraints [2].

Meanwhile, the cosmological horizon is a 2-sphere with area $4\pi R_H^2$. The spherical boundary introduces factor π. The ratio $k/\pi = 4/\pi$ represents the mismatch between:

- **Local flat frame:** $k = 4$ tetrad constraint structure
- **Curved global boundary:** π spherical cosmological horizon

In Regge calculus [5], curved spacetime is approximated by a patchwork of flat simplices (tetrahedra), with curvature encoded in "deficit angles" at the edges. The same principle applies here: measurements traverse from flat local frames to a curved global boundary, accumulating frame corrections.

Since the Hubble constant $H = v/d$ depends on length (not area), and the k/π ratio is an area measure, we take the square root:

$$H_{\text{measured}} = \frac{H_{\text{flat}}}{\sqrt{k/\pi}} = \frac{78.90}{1.128} = 69.92 \text{ km/s/Mpc} \tag{29}$$

B. Why $\pm 6.2\%$ Scatter?

Different measurement methods integrate through different portions of the frame transition from local flat frames to curved global boundary:

- **Early-universe methods (Planck, BAO):** Integrate from cosmological distances, accumulating more of the curved-boundary correction. Result: lower H values (67–68 km/s/Mpc).
- **Late-universe methods (SH0ES, TRGB, Masers):** Integrate through nearby structure where the local flat frame dominates. Result: higher H values (72–74 km/s/Mpc).
- **Direct spacetime measurement (GW170817):** Gravitational waves propagate through the metric itself, sampling an intermediate path through the frame structure. Result: $H \approx 70$ km/s/Mpc.

Topology defines the expected scatter. When we observe the universe, each measurement traces a path through the $k = 4$ field topology. The topology itself determines how much scatter we should expect. The full frame correction is $\sqrt{k/\pi} = 1.128$ (12.8%), but individual measurements do not experience this correction uniformly. They experience *partial* corrections depending on their observational path:

- A measurement entirely within the flat local frame experiences 0% correction.
- A measurement that fully samples the curved boundary experiences 12.8% correction.
- Most measurements fall somewhere between, depending on how much of the frame transition they traverse.

The expected scatter is the geometric mean of partial corrections. For a multiplicative correction factor f ranging uniformly from 1 to $\sqrt{k/\pi}$ on a logarithmic scale, the geometric mean is:

$$\text{Scatter factor} = \sqrt{1 \times \sqrt{k/\pi}} = (k/\pi)^{1/4} = 1.062 \quad (\pm 6.2\%) \tag{30}$$

Why the geometric mean? The frame correction is multiplicative, not additive: $H_{\text{measured}} = H_{\text{flat}}/f$ where f ranges from 1 to 1.128. For multiplicative processes, the appropriate average is the geometric mean, not the arithmetic mean. The geometric mean of f ranging from 1 to $\sqrt{k/\pi}$ is $\sqrt{1 \times 1.128} = 1.062$. The fourth root arises because we take the geometric mean (square root) of a quantity that is itself a square root ($\sqrt{k/\pi}$).

Note: We do not claim to have derived the full statistical distribution of H measurements. What we claim is more modest: the $k = 4$ topology defines a *characteristic scale* for path-dependent scatter. Individual measurements, sampling different paths through the topology, should cluster within this scale. The observed data are consistent with this prediction: all six major methods fall within the ±6.2% range, despite using fundamentally different physics and probing different cosmic epochs.

This defines a "measurement cage" spanning 65.8–74.3 km/s/Mpc:

$$\begin{aligned} \text{Lower bound:} \quad & 69.92/1.062 = 65.8 \text{ km/s/Mpc} \\ \text{Upper bound:} \quad & 69.92 \times 1.062 = 74.3 \text{ km/s/Mpc} \end{aligned}$$

The framework predicts all measurements should fall within this cage:

Method	**H_0 (km/s/Mpc)**	**Deviation**
Planck (CMB) [8]	67.4	−3.6%
BAO	68.0	−2.8%
GW170817	**70.0**	**+0.11%**
TRGB	72.5	+3.7%
SH0ES [9]	73.0	+4.4%
Masers	73.9	+5.7%

TABLE IV. All six major H_0 measurements fall within the predicted ±6.2% cage (65.8–74.3 km/s/Mpc).

All six major methods fall within the predicted cage. The cage width (±6.2%) spans 12.4% of the plausible range (50–150 km/s/Mpc). If measurements were uniformly distributed across this range, the probability of all six landing within a 12% window would be $(0.12)^6 \approx 3.0\times10^{-6}$. (In practice, measurements share some systematics, so this overstates significance; the qualitative point is that the cage is predictive, not fitted.)

Convergence test: Earlier, we divided by $\sqrt{k/\pi} = 1.128$ to convert the MOND equation value (78.90) to the measurable value (69.92). We can run this in reverse: take each of the six observed H values and multiply by 1.128. If our framework is correct, they should all converge to 78.90. They do; the average is 79.1 km/s/Mpc (0.27% match), and GW170817 alone gives 79.0 km/s/Mpc (0.11% match). Six measurements that look different all point to the same underlying value once the topology is accounted for.

The "tension" between 67 and 73 km/s/Mpc is not a crisis; it is the topological signature of measuring through $k = 4$ structure to a spherical boundary.

C. Falsifiability

A referee may ask: "What would disprove this framework?" The theory makes specific predictions that could fail:

1. **The coupling ratio.** We predict $a_0/(Gm_e/r_e^2) = k^2 = 16$. The measured value is 15.93. If improved measurements of a_0 or electron parameters shifted this ratio outside 16 ± 1, the framework would be falsified.

2. **The H ratio.** We predict $H_{\text{flat}}/H_{\text{measured}} = \sqrt{k/\pi} = 1.128$. If the GW170817-type measurements (which bypass the distance ladder) converged on a value outside 70 ± 3 km/s/Mpc, the framework would be falsified.

3. **The measurement cage.** We predict all H measurements should fall within $(k/\pi)^{1/4} = \pm 6.2\%$ of the predicted value, i.e., 65.8–74.3 km/s/Mpc. A credible measurement outside this range would challenge the framework. (Note: this has held for all 7 major methods tested.)

4. **The electron mass prediction.** Given H, we predict m_e to 0.10%. Future precision measurements of either H or m_e that break this correspondence would falsify the framework.

Critically, the value $k = 4$ is not adjustable; it is fixed by the simplex theorem. If data required $k = 5$ or $k = 3$, no adjustment could save the theory.

D. Discussion

Why does this work? Gravity is a field. Fields that propagate must couple to neighbors. Coupling requires consistency: where two field configurations meet, they must agree. This consistency requirement is field closure. Field closure in 3+1 spacetime requires $k = 4$ topology. The topology is not imposed; it emerges from the requirement that distributed fields settle into consistent states.

A fish does not see the water molecules it swims through, but the molecules determine how waves propagate. We do not see the gravitational field structure we measure through, but the structure determines how cosmological observables appear. The framework works because it accounts for what measurements pass through: the $k = 4$ topology of gravitational field closure.

The tile size question. The tetrad formalism specifies topology ($k = 4$ faces) but not scale. Three independent routes converge on $r_e = 2.82$ fm: electromagnetic stability, cosmic

topology, and MOND prediction. This convergence raises a question: is r_e merely where our measurement happens to land, or is it fundamental to the gravitational field itself?

Consider the analogy of ocean waves. Waves propagate because the ocean consists of molecules, each pushing the next. We do not derive the size of a water molecule from wave dynamics; we observe it independently. But the wave behavior is *consistent* with the molecular structure. Similarly, we do not derive r_e from topology alone. We observe that topology combined with cosmological constraints converges on the scale where electromagnetism stabilizes. The consistency across independent physics suggests r_e may be the characteristic coupling scale of gravity: the "tile size" of the topological structure.

Bidirectional convergence as evidence. The electron scale emerges from two completely independent calculations: electromagnetic stability ($r_e = e^2/4\pi\varepsilon_0 m_e c^2 = 2.818$ fm) and gravitational topology ($r_e = k\sqrt{Gm_e/a_0} = 2.82$ fm). These routes use different physics (one has no gravity, the other no electromagnetism) yet converge to 0.21%. This convergence is unique to the electron; for the muon and proton, the two routes disagree by factors of 10^3–10^7.

This bidirectional convergence is significant. If r_e were merely a convenient mathematical choice, there is no reason two unrelated domains of physics would agree on its value. The fact that electromagnetism and gravity-topology both independently identify $r_e = 2.82$ fm, and that this convergence is *unique* to the electron among known particles, suggests this scale is not arbitrary. Combined with the 0.11% match to observed H when the lens correction is applied at this scale, the evidence points toward r_e being the actual tile size of the gravitational field's topological structure. We do not claim this as proven; we note it as the implication of the convergence.

What this framework does not explain. Two questions remain open:

1. **Why m_e?** The electron mass enters as input. A complete theory would derive m_e from topological constraints. The 0.10% round-trip closure ($m_e \to H \to m_e$) suggests this may be possible, but it is not demonstrated here.

2. **Why a_0?** We use $a_0 = cH/(2\pi)$ from horizon thermodynamics as a boundary condition. A deeper theory would derive this relation from the topology itself.

The methodological point. Standard approaches seek new physics to resolve the Hubble

tension. This framework asks a different question: what if the tension is not a problem but a prediction? The $k = 4$ topology has been in ADM constraints and tetrad formalism for sixty years. We did not invent it; we applied it to cosmology. The result: the scatter between 67 and 73 km/s/Mpc is the predicted signature of measuring through $k = 4$ flat local frames to a curved cosmological boundary.

V. CONCLUSION

The problem. The Hubble tension describes the persistent discrepancy between early-universe measurements ($H \approx 67$ km/s/Mpc) and late-universe measurements ($H \approx 73$ km/s/Mpc). This paper applied the tetrad formalism of General Relativity. The tetrad requires $k = 4$ orthonormal basis vectors to close the local frame. Field closure with $k = 4$ topology is the dimensional requirement of 3+1 spacetime.

What we identified. Two independent routes identify the minimal closure size of the gravitational field: 2.82 fm. Electromagnetic stability gives $r_e = e^2/4\pi\varepsilon_0 m_e c^2 = 2.818$ fm; gravitational topology gives $r_e = k\sqrt{Gm_e/a_0} = 2.82$ fm. These routes use different physics yet converge to 0.21%.

How we resolved it. The MOND equation $a_0 = cH/(2\pi)$ predicts $H = 78.90$ km/s/Mpc. Yet a century of observations has measured H between 67 and 73, never near 79. The tetrad formalism resolves this discrepancy: gravity has field structure, and measurements pass through it. Projecting MOND through the $k = 4$ tetrad topology gives $H = 78.90/\sqrt{k/\pi} = 69.92$ km/s/Mpc, matching GW170817 (70.0 ± 12) to 0.11%. The derivation is bidirectional: the observed H recovers the electron mass to 0.10% of CODATA.

The tension explained. The framework predicts all Hubble measurements scatter within $\pm 6.2\%$ of 69.92 km/s/Mpc. All six major methods fall within this range. The apparent tension between early and late-universe values is the predicted consequence of measuring through $k = 4$ flat local frames to a curved cosmological boundary. Three independent tests confirm $k = 4$.

Limitations. This framework does not derive the electron mass from first principles; m_e enters as input. The MOND equation is taken from horizon thermodynamics, not derived from the $k = 4$ topology. The frame correction $\sqrt{k/\pi}$ is a dimensional argument, not a

rigorous derivation from Regge calculus.

Falsifiability. All Hubble measurements should fall within 65.8–74.3 km/s/Mpc; a credible measurement outside this range would challenge the framework. The value $k = 4$ is not adjustable; it is fixed by the dimensional requirement of 3+1 spacetime.

DATA AVAILABILITY

A Python script that independently verifies all numerical calculations in this paper is available as supplementary material. The script requires only Python 3.6+ with no external dependencies. All input values are CODATA 2022 fundamental constants; no fitted parameters are used.

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